LAB ACTIVITY · GRASP FOR THE TSP

Does biased randomization make GRASP better? A test on the travelling salesman problem

Clarke-Wright savings, four selection rules and TSPLIB benchmarks

Javier Salvatierra Corchado

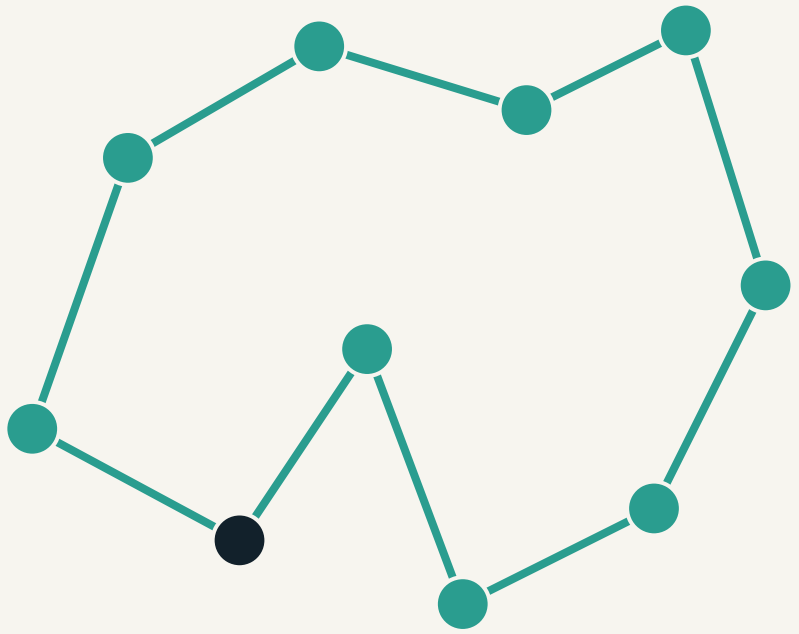
The travelling salesman problem

Visit every city exactly once and return to the start.

Minimise the total length of the tour.

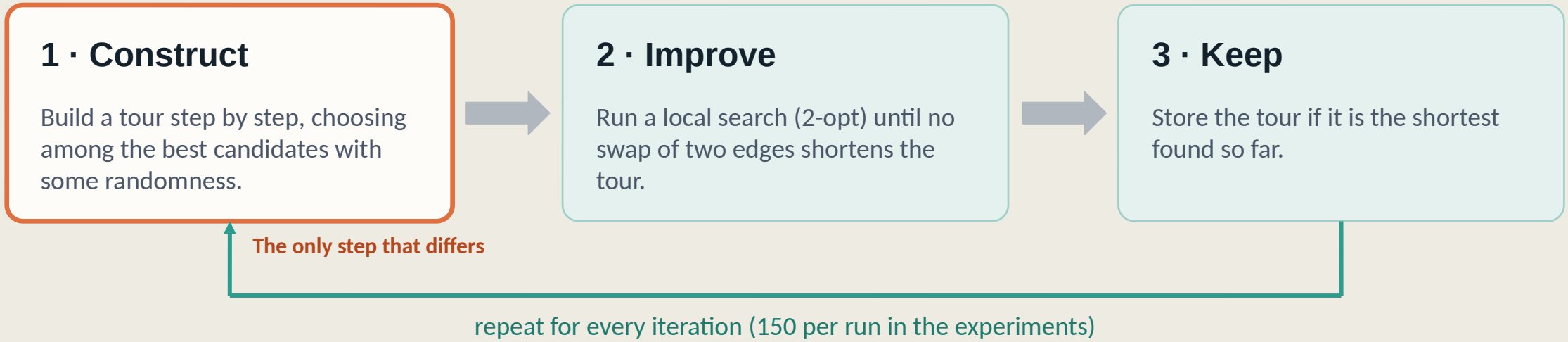
NP-hard: the number of possible tours grows factorially with the number of cities.

How close can a fast heuristic get to the optimum?



Benchmark: TSPLIB, 7 test instances (51 to 280 cities) with known optima. Metric: gap = (tour cost / optimum − 1) × 100.

GRASP is one loop: build, improve, keep the best



GREEDY

Candidates are ranked by a score; here, the saving of joining two cities.

RANDOMIZED

The pick is not always the top one, so each iteration builds a different tour.

ADAPTIVE

After each pick, the list of valid candidates is updated.

Only the choice rule in step 1 differs between the methods we compare.

GRASP: Greedy Randomized Adaptive Search Procedure. The local search is 2-opt, first improvement, in every variant.

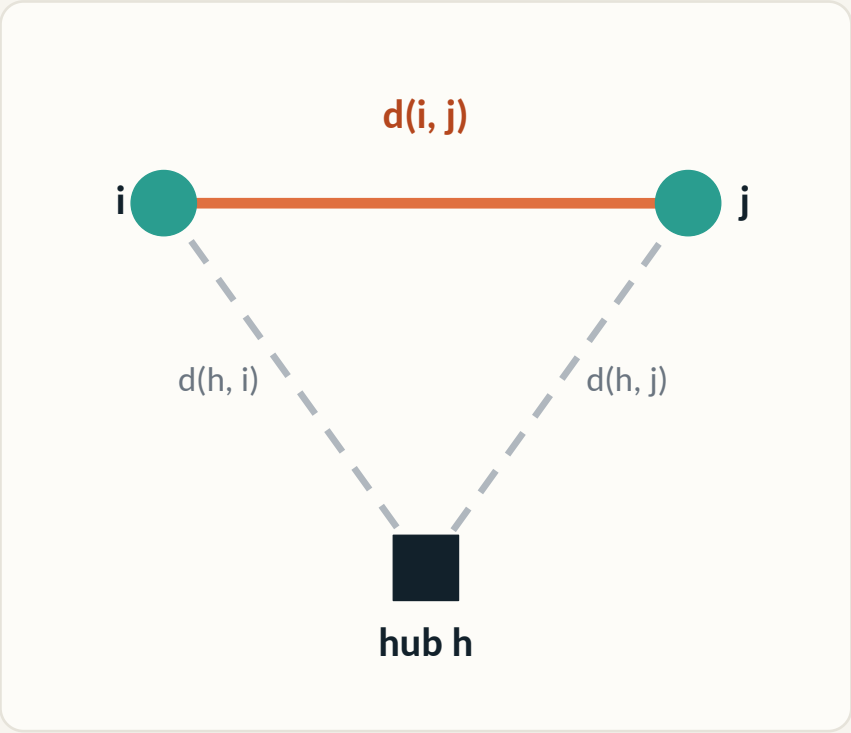
Clarke-Wright savings rank the candidates

$$s(i, j) = d(h, i) + d(h, j) - d(i, j)$$

h is the hub: the city closest to the centre of the instance.

i, j are any two other cities.

s(i, j) is the distance saved by travelling i → j directly instead of going back to the hub in between.



A large saving makes a pair attractive to join.

The savings of all pairs are computed once per instance and sorted from largest to smallest. Clarke and Wright (1964).

Building one tour: filter, choose, merge

STEP 1

Filter

Keep only pairs (i, j) where both cities are ends of their route and the routes are different. Anything else would create a branch or a cycle.

STEP 2 · WHERE METHODS DIFFER

Choose

Pick one pair from the ranked list. The classic GRASP uses an RCL; biased randomization uses a probability that decays with the rank.

STEP 3

Merge

Join the two routes with the chosen pair and go back to step 1. After n − 2 merges one path is left, closed through the hub.

WORKED EXAMPLE · 4 CITIES

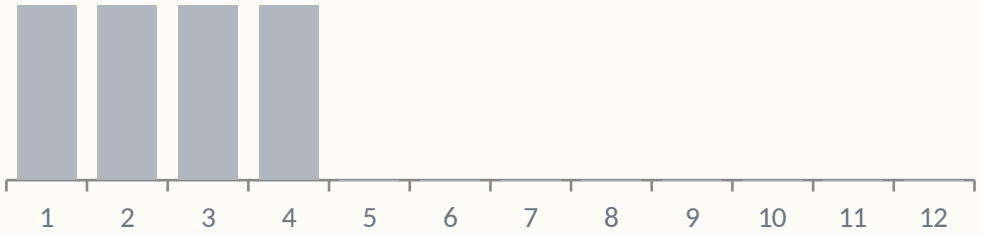
Join A–B, then B–C, then C–D. The path A–B–C–D, closed through the hub, is the tour.

The candidate list shrinks after every merge: this is the adaptive part of GRASP.

Two ways to pick the next pair

Classic GRASP: the RCL

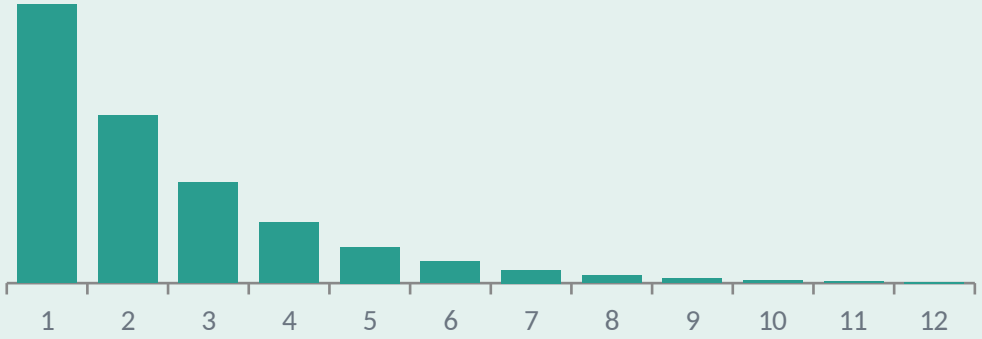
Keep the pairs whose saving is within a fraction α of the best one. Pick one of them with equal probability.



Rank in the list, best saving on the left. Uniform inside, zero outside.

Biased randomization

Rank every feasible pair by saving. Rank k is picked with a probability that decays as k grows. No cut-off: every pair keeps a chance.



Rank in the list, best saving on the left. Smooth decay, long tail.

Same ranked list, same local search: only the probability rule changes.

Illustrative shapes: an RCL with 4 pairs and a geometric rule with β = 0.4, drawn on the same probability scale.

Four ways to bias the choice

RULE	PROBABILITY OF PICKING RANK k	PARAMETER	LOOKS AT
Geometric	∝ β (1 − β)^k	β	the rank
Triangular	linear decay over the top ⌈f · m⌉ ranks	f	the rank
Zipf	∝ 1 / (k + 1)^e	e	the rank
Softmax	∝ exp(z_k / T), z_k = normalised saving	T	**the saving value**

Three rules use only the rank. Softmax also sees how much better one saving is than another.

m = number of feasible pairs; k = rank, 0 for the best saving. Geometric is the usual choice in the biased-randomization literature.

A fair comparison: only the choice rule changes

Held constant

- Clarke-Wright construction, same hub
- 2-opt local search, first improvement
- 150 iterations per run, same seeds

Tuned fairly

- Every rule gets a grid of 6 to 8 values
- Tuned on kroB100, eil101 and ch150, never on the test set
- Best value = lowest mean gap

Tested

- 7 instances, 51 to 280 cities
- 20 runs per instance and rule
- Wilcoxon rank-sum with Holm correction; Friedman across instances

TUNED PARAMETERS

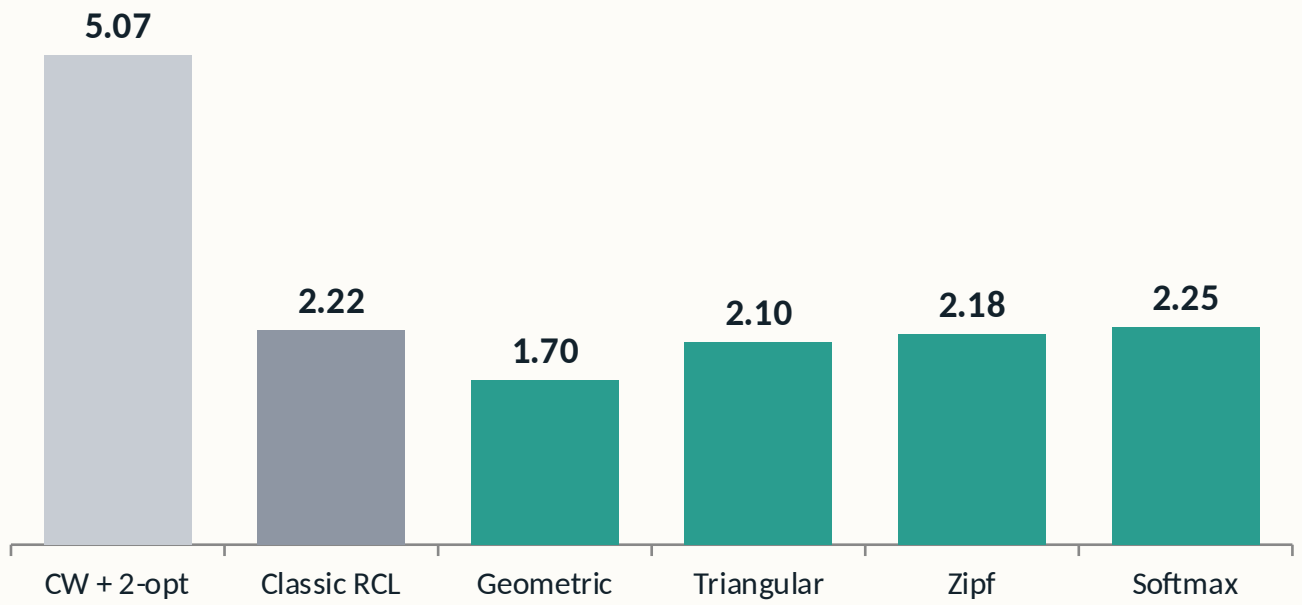
RCL · α = 0.05 | **Geometric · β = 0.4** | **Triangular · f = 0.002** | **Zipf · e = 3** | **Softmax · T = 0.02**

Any difference comes only from how the next pair is chosen.

Test instances: eil51, berlin52, st70, eil76, kroA100, ch130, a280.

Quality and time, by selection rule

AVERAGE GAP TO THE OPTIMUM, % · LOWER IS BETTER



SECONDS PER RUN ON a280

Classic RCL	5.8 s
Geometric	5.3 s
Triangular	5.5 s
Zipf	10.2 s
Softmax	9.9 s

GRASP cuts the error of deterministic Clarke-Wright from 5.07% to 2.22%. The geometric rule lowers it a further 23%, at no extra cost.

Mean over 7 TSPLIB instances, 20 runs × 150 iterations. CW + 2-opt: deterministic Clarke-Wright followed by 2-opt.

Where each rule wins, instance by instance

Gap, %	Classic RCL	Geometric	Triangular	Zipf	Softmax
eil51	1.41	1.10	**0.95 ▲**	1.20	1.41
berlin52	3.69	**0.93 ▲**	**2.04 ▲**	3.35	3.44
st70	0.82	1.01	0.99	**1.26 ▼**	0.93
eil76	1.78	1.65	1.66	2.02	1.89
kroA100	0.94	1.14	1.12	1.19	0.91
ch130	1.71	1.73	1.80	1.54	1.46
a280	5.16	**4.31 ▲**	**6.14 ▼**	4.74	5.70
Mean	**2.22**	**1.70**	**2.10**	**2.18**	**2.25**
▲ / = / ▼	baseline	**2 / 5 / 0**	**2 / 4 / 1**	**0 / 6 / 1**	**0 / 7 / 0**

Reading the table

Geometric: better on berlin52 and a280, never worse.

Triangular: better on eil51 and berlin52, worse on a280.

Zipf, softmax: no clear gain over the classic RCL.

Friedman across instances: p = 0.76. Moderate evidence, not conclusive.

▲ significantly better than the classic RCL, ▼ significantly worse (Wilcoxon rank-sum, Holm correction, 5%). Mean gap in %, 20 runs × 150 iterations.

TAKEAWAY · OUR READING OF THE RESULTS

With savings, the geometric rule gave the lowest gap, but the evidence is moderate.

WORTH ASKING ABOUT

Moderate evidence: two significant wins for the geometric rule, and Friedman p = 0.76 across the 7 instances.

Parameters were tuned on three instances with few runs; more tuning data could change the ranking.

2-opt is a basic local search; Or-opt or neighbour lists could shrink the differences.

NEXT STEPS

Larger instances and more runs, to firm up the statistics.

Equal-CPU-time comparison: Zipf and softmax take roughly 1.7 times longer per run.

Path relinking or an elite set, to exploit the diversity the bias creates.

Thank you — questions?